

Web and Multimedia Technologies

Lab class 1 – GIMP

Instructions to use the laboratory computers (rooms B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the lesson, please logout

Steps (related to GIMP, <https://www.gimp.org/>):

1. Create the folder *gimp* within your personal folder (or where you prefer)
2. Download the file <https://vision.unipv.it/wmt/lab/bear.jpg> to the *gimp* folder
3. Open GIMP and load *bear.jpg* (*Open...* command from the *File* menu). Throughout the following steps, every now and then, save the image (in *xcf* format, that is the native format of GIMP): *File* → *Save* or CTRL + S (⌘-S on Mac)
4. Select/deselect the RGB channels in the *Channels* dialog (on the right) and observe the results
5. Improve image quality by modifying levels (*Colors* → *Levels...*); then undo the operation with CTRL + Z
6. Modify the image by adjusting *curves* (*Colors* → *Curves...*)
7. In the *Undo History* dialog (on the left), undo the operation performed in the previous step and then improve again the quality of the image by modifying levels (step 5)
8. Modify the image by adjusting color balance (*Colors* → *Color Balance...*)
9. After undoing the operation performed in the previous step, modify the image by adjusting brightness and contrast (*Colors* → *Brightness-Contrast...*)
10. After undoing the operation of the previous step, modify the image by adjusting hue and saturation (*Colors* → *Hue-Saturation...*)
11. Undo the previous operation
12. Using the *Rectangle Select Tool* (panel in the upper left corner), select the bear's head. Copy the selection (*Edit* → *Copy* or CTRL + C on the keyboard) and create a new image the same size as the copied selection (*Edit* → *Paste as* → *New Image* or SHIFT + CTRL + V)
13. Go back to the original image (*bear.jpg*) and remove the selection by clicking on any point outside it
14. Using the *Free Select Tool*, select the bear's head (this time moving the mouse cursor along its contour). Move the head to where you prefer (*Select* → *Float*)
15. Using again the *Free Select Tool* (or any other selection tool), click anywhere (outside the selected area) to remove the selection
16. Choose a foreground color in the *Change Foreground Color* window
17. Select the "hole" left by the bear's head using the *Fuzzy Select Tool* and fill it using the *Bucket Fill Tool* (the properties of a tool can be set in the *Tool Options* dialog)
18. With the *Painbrush Tool* (after selecting proper parameters in its toolbar), draw eyes, nose and mouth in the area formerly occupied by the bear's head
19. With the *Eraser Tool*, erase some areas of the image

20. With the *Blur/Sharpen Tool*, blur the “actual” mouth of the bear (wherever its head is...)
21. With the *Text Tool*, write “Brown Bear” somewhere in the image and move it where you want (*Move Tool*, active layer)
22. In the *Layers* dialog (bottom right), click on the background layer (“bear.jpg”), to select it
23. Draw (where you want in the image): straight lines (select a drawing tool and click where you want the line to start; then hold down the SHIFT key and click where you want the line to end); rectangles (create selections with the *Rectangle Select Tool* and fill them with the *Bucket Fill Tool*); ellipses (create selections with the *Ellipse Select Tool* and fill them with the *Bucket Fill Tool*)
24. Create a new layer (leftmost button at the bottom of the *Layers* dialog on the right)
25. With the newly created layer selected, use the *Paths Tool* to move along the outline of the (actual) head of the bear (each time you click, a new node is created); the path will be listed in the *Paths* dialog (bottom right)
26. In the *Paths* dialog, right click on the path and select *Stroke Path...* to choose stroke parameters and press the *Stroke* button to draw on the path
27. In the *Layers* dialog, select the “bear.jpg” layer
28. With the *Clone Tool*, replicate some areas of the image in other zones; keep the CTRL key pressed and click to choose the area to be reproduced (by moving the mouse cursor around with the left button pressed)
29. Contemplate the just created artwork (and sincerely apologize to the poor bear...)
30. Save the image in JPEG format as “bear1.jpg” (*File* → *Export As...*)
31. Convert the image to *Indexed Color* (i.e., 8 bits, 256 colors at most): *Image* → *Mode* → *Indexed...*
32. Create an Alpha channel for transparency: *Layer* → *Transparency* → *Add Alpha Channel*
33. Using a selection tool, select some areas of your choice in the image and press the DEL key on the keyboard, or CTRL-X, or select *Edit*→*Cut*, to create transparent areas
34. Save the image in GIF format as “bear1.gif” (*File* → *Export As...*)
35. Reopen the original file `bear.jpg` (if you closed it)
36. Apply some filters of your choice (*Filters* menu)